ClassNest

UNIT-V Advanced Python Integration and Data Handling Assessment

UNIT V: C++ Integration, SQLite Data Manipulation and Data Visualization

Coverage: Importing C++ Programs in Python, Data Manipulation through SQL, Data Visualization using PyPlot
Pattern: 30 MCQs, 20 Fill-ups, 10 Simple Coding Questions. Total: 60 questions, 100 marks.
Student paper. Answers are intentionally hidden.

MCQ QUESTIONS

QUESTION 1 - MCQ - 1 MARK

In the Python-C++ comparison, Python is typically described as a/an:

A. Compiled language
B. Interpreted language
C. Assembly language
D. Markup language

QUESTION 2 - MCQ - 1 MARK

Which language is usually statically typed according to the C++ comparison?

A. Python
B. C++
C. HTML
D. CSV

QUESTION 3 - MCQ - 1 MARK

Python can act as a scripting language mainly because it can:

A. Only draw charts
B. Integrate and communicate with other languages
C. Only create databases
D. Only compile C++

QUESTION 4 - MCQ - 1 MARK

Which of the following is an application of scripting languages?

A. Automating tasks
B. Deleting operating systems
C. Replacing hardware
D. Changing monitor size

QUESTION 5 - MCQ - 1 MARK

Python uses automatic garbage collection, whereas C++:

A. Does not use automatic garbage collection in the same way
B. Cannot use variables
C. Cannot use functions
D. Is not a programming language

QUESTION 6 - MCQ - 1 MARK

Importing a C++ program in a Python program is called:

A. Plotting
B. Wrapping
C. Sorting
D. Indexing

QUESTION 7 - MCQ - 1 MARK

Which interface is commonly used for interfacing Python with C programs?

A. Python-C-API
B. CSV module
C. PyPlot only
D. fetchall()

QUESTION 8 - MCQ - 1 MARK
Which tool is expanded as Simplified Wrapper Interface Generator?

A. SQLite
B. SWIG
C. MinGW
D. PyPlot

QUESTION 9 - MCQ - 1 MARK
MinGW is mainly used for:

A. Compiling and linking C/C++ programs on Windows
B. Creating pie charts only
C. Opening CSV files only
D. Writing HTML pages

QUESTION 10 - MCQ - 1 MARK
In the command `python pycpp.py -i pali`, the `-i` part represents:

A. input mode
B. integer mode
C. image mode
D. insert mode

QUESTION 11 - MCQ - 1 MARK
The Python library used in the notes to work with SQLite is:

A. sqlite3
B. matplotlib
C. csvonly
D. numpydb

QUESTION 12 - MCQ - 1 MARK
SQLite stores its data in:

A. Regular database files
B. Only printed paper
C. Only cloud servers
D. Image files only

QUESTION 13 - MCQ - 1 MARK
Which method is used to create a connection to a SQLite database?

A. connect()
B. cursor() only
C. fetchall()
D. show()

QUESTION 14 - MCQ - 1 MARK
The cursor object is created by calling:

A. connection.cursor()
B. database.paint()
C. plot.cursor()
D. csv.cursor()

QUESTION 15 - MCQ - 1 MARK
In SQLite/Python, SQL commands are executed using:

A. cursor.execute()
B. list.append()
C. plt.plot()
D. field.split()

QUESTION 16 - MCQ - 1 MARK
After INSERT/UPDATE/DELETE operations, changes should be saved using:

A. commit()
B. fetchone()
C. show()
D. ylabel()

QUESTION 17 - MCQ - 1 MARK

A cursor is mainly used to:

A. Traverse and fetch database records
B. Make a pie chart
C. Compile C++ only
D. Store CSV formulas

QUESTION 18 - MCQ - 1 MARK

Which SQL command is used to retrieve records from a table?

A. SELECT
B. CREATE only
C. DROP only
D. COMMIT only

QUESTION 19 - MCQ - 1 MARK

fetchall() is used to:

A. Fetch all rows from the result set
B. Fetch no rows
C. Delete a table
D. Create a plot title

QUESTION 20 - MCQ - 1 MARK

fetchone() returns:

A. The next row or None if no row is left
B. All tables in database
C. A bar chart
D. A C++ compiler

QUESTION 21 - MCQ - 1 MARK

A data visualization is best described as:

A. Graphical representation of data/information
B. Raw data without meaning
C. A database file only
D. A C++ header file

QUESTION 22 - MCQ - 1 MARK

Which of the following is listed as a type of data visualization?

A. Charts
B. Compilers
C. Pointers only
D. Primary keys only

QUESTION 23 - MCQ - 1 MARK

A dashboard is mainly:

A. A unified visual display of resources/information
B. A Python compiler
C. A database cursor
D. A C++ input mode

QUESTION 24 - MCQ - 1 MARK

Matplotlib is a popular Python library for:

A. Data visualization
B. C++ compilation only
C. Database locking only
D. CSV file extension

QUESTION 25 - MCQ - 1 MARK

Which plot displays data as a collection of points?

A. Scatter plot
B. Pie chart
C. Transaction log
D. SQL table

QUESTION 26 - MCQ - 1 MARK
A box plot summarizes distribution based on:

A. Five-number summary
B. Only one number
C. Only file name
D. Only column color

QUESTION 27 - MCQ - 1 MARK
Which chart is suitable to show parts of a whole?

A. Pie chart
B. Compiler chart
C. Cursor chart
D. Header chart

QUESTION 28 - MCQ - 1 MARK
Which chart commonly compares values across categories?

A. Bar chart
B. GCC chart
C. SQL commit chart
D. CSV quote chart

QUESTION 29 - MCQ - 1 MARK
Which function is commonly used to display a Matplotlib chart window?

A. show()
B. commit()
C. execute()
D. cursor()

QUESTION 30 - MCQ - 1 MARK
Which method is used to label the x-axis in PyPlot?

A. xlabel()
B. connect()
C. fetchone()
D. wrap()

FILLUP QUESTIONS

QUESTION 31 - FILLUP - 1 MARK
Python is typically an ______ language, while C++ is typically compiled.

Answer: ________________________________

QUESTION 32 - FILLUP - 1 MARK
C++ is usually ______ typed, while Python is dynamically typed.

Answer: ________________________________

QUESTION 33 - FILLUP - 1 MARK
The process of importing C++ code into Python is called ______.

Answer: ________________________________

QUESTION 34 - FILLUP - 1 MARK
SWIG stands for Simplified Wrapper Interface ______.

Answer: ________________________________

QUESTION 35 - FILLUP - 1 MARK
MinGW allows C++ programs to be compiled and executed dynamically through a ______ program.

Answer: ________________________________

QUESTION 36 - FILLUP - 1 MARK
The Python module used for SQLite database handling is ______.

Answer: ________________________________

QUESTION 37 - FILLUP - 1 MARK
The sqlite3 ______ method opens or creates a database file.

Answer: ______________________________

QUESTION 38 - FILLUP - 1 MARK
The cursor object is created using the ______ method of a connection.

Answer: ______________________________

QUESTION 39 - FILLUP - 1 MARK
The SQL command is run in Python using cursor.______().

Answer: ______________________________

QUESTION 40 - FILLUP - 1 MARK
Changes made to database records are saved permanently using ______().

Answer: ______________________________

QUESTION 41 - FILLUP - 1 MARK
The SELECT statement is used to ______ data from a table.

Answer: ______________________________

QUESTION 42 - FILLUP - 1 MARK
The ______ method fetches all rows from a query result.

Answer: ______________________________

QUESTION 43 - FILLUP - 1 MARK
The fetchone() method returns ______ if there is no row left.

Answer: ______________________________

QUESTION 44 - FILLUP - 1 MARK
Data visualization is the graphical representation of ______ and data.

Answer: ______________________________

QUESTION 45 - FILLUP - 1 MARK
Matplotlib is a popular data visualization ______ in Python.

Answer: ______________________________

QUESTION 46 - FILLUP - 1 MARK
A scatter plot shows data as a collection of ______.

Answer: ______________________________

QUESTION 47 - FILLUP - 1 MARK
A box plot displays a distribution based on the ______ number summary.

Answer: ______________________________

QUESTION 48 - FILLUP - 1 MARK
The middle value in a box plot five-number summary is the ______.

Answer: ______________________________

QUESTION 49 - FILLUP - 1 MARK
A dashboard translates complex ideas into a simple ______ format.

Answer: ______________________________

QUESTION 50 - FILLUP - 1 MARK
The PyPlot function ______() is commonly used to display the graph.

Answer: ______________________________

CODING QUESTIONS

QUESTION 51 - CODING - 5 MARKS

Write a function build_cpp_command(py_file, cpp_name) that returns the command string to execute a Python wrapper with input mode. Example: build_cpp_command("pycpp.py", "pali") should return "python pycpp.py -i pali".

Starter Code:

```
def build_cpp_command(py_file, cpp_name):
    # write your code here
    pass
```

Visible Test Cases:

```
Input: build_cpp_command("pycpp.py", "pali")
Expected: python pycpp.py -i pali

Input: build_cpp_command("run.py", "demo")
Expected: python run.py -i demo
```

QUESTION 52 - CODING - 5 MARKS

Write a function is_palindrome_number(n) that returns True if the integer reads the same forward and backward, otherwise False.

Starter Code:

```
def is_palindrome_number(n):
    # write your code here
    pass
```

Visible Test Cases:

```
Input: is_palindrome_number(232)
Expected: True

Input: is_palindrome_number(234)
Expected: False
```

QUESTION 53 - CODING - 5 MARKS

Write a function sqlite_connect_line(db_name) that returns the Python code line for creating a SQLite connection. Example: sqlite_connect_line("Academy.db") returns "connection = sqlite3.connect("Academy.db")".

Starter Code:

```
def sqlite_connect_line(db_name):
    # write your code here
    pass
```

Visible Test Cases:

```
Input: sqlite_connect_line("Academy.db")
Expected: connection = sqlite3.connect("Academy.db")
```

QUESTION 54 - CODING - 5 MARKS

Write a function make_insert_student(name, grade, gender, average, birth_date) that returns an SQL INSERT string using NULL for Rollno.

Starter Code:

```
def make_insert_student(name, grade, gender, average, birth_date):
    # write your code here
    pass
```

Visible Test Cases:

```
Input: make_insert_student("Akshay", "B", "M", 87.8, "2001-12-12")
Expected: INSERT INTO Student (Rollno,Sname,Grade,gender,Average,birth_date) VALUES
    (NULL,"Akshay","B","M",87.8,"2001-12-12");
```

QUESTION 55 - CODING - 5 MARKS

Write a function filter_above_average(records, limit) that receives a list of tuples (name, average) and returns the names whose average is greater than limit.

Starter Code:

```
def filter_above_average(records, limit):
    # write your code here
    pass
```

Visible Test Cases:

```
Input: filter_above_average([("Baskar",75.2),("Priya",98.6)], 80)
Expected: ["Priya"]
```

QUESTION 56 - CODING - 5 MARKS

Write a function fetch_one_simulated(rows) that returns the first row from a list, or None if the list is empty.

Starter Code:

```
def fetch_one_simulated(rows):
    # write your code here
    pass
```

Visible Test Cases:

```
Input: fetch_one_simulated([(1,"Akshay"),(2,"Aravind")])
Expected: (1,"Akshay")

Input: fetch_one_simulated([])
Expected: None
```

QUESTION 57 - CODING - 5 MARKS

Write a function count_records(rows) that returns the number of records returned by a query result list.

Starter Code:

```
def count_records(rows):
    # write your code here
    pass
```

Visible Test Cases:

```
Input: count_records([(1,"A"),(2,"B"),(3,"C")])
Expected: 3
```

QUESTION 58 - CODING - 5 MARKS

Write a function choose_chart_type(purpose) that returns "bar" for comparison, "pie" for parts, "line" for trend, and "scatter" for relationship. Return "unknown" otherwise.

Starter Code:

```
def choose_chart_type(purpose):
    # write your code here
    pass
```

Visible Test Cases:

```
Input: choose_chart_type("comparison")
Expected: bar

Input: choose_chart_type("parts")
Expected: pie
```

QUESTION 59 - CODING - 5 MARKS

Write a function five_number_summary(values) that returns a tuple (minimum, q1, median, q3, maximum). For this assessment, assume the input list has exactly five sorted values.

Starter Code:

```
def five_number_summary(values):
    # values will contain exactly five sorted numbers
    # write your code here
    pass
```

Visible Test Cases:

```
Input: five_number_summary([10,20,30,40,50])
Expected: (10,20,30,40,50)
```

QUESTION 60 - CODING - 5 MARKS

Write a function pyplot_line_code(x_values, y_values) that returns the code string "plt.plot(x_values, y_values)". This checks your understanding of basic PyPlot line plotting syntax.

Starter Code:

```
def pyplot_line_code(x_values, y_values):
    # write your code here
    pass
```

Visible Test Cases:

```
Input: pyplot_line_code([1,2,3], [2,4,6])
Expected: plt.plot(x_values, y_values)
```